

ARGUSTREAM: Live Joint Argument–Entity Analysis of Political Debates

Anonymous submission

Abstract

Argument Mining (AM) has largely focused on retrospective analysis of completed argumentative texts. ARGUSTREAM, a new interactive system for turn-synchronous joint argument-entity analysis of political debates brings argument analysis live. As a debate unfolds, ARGUSTREAM transforms each speaker turn into a structured representation of *who* is arguing, *what* they are arguing, and *which entities* their arguments concern. The system jointly identifies argumentative components and debate-relevant entities in a single inference pass, and integrates them into a continuously evolving debate state that can be explored by speaker, entity, argumentative roles, and time. ARGUSTREAM supports both *real-time* and *post-hoc analysis* and paves the way toward interactive, real-time understanding of unfolding argumentative discourses.

Introduction

Argument Mining (AM) aims to automatically identify the argumentative structure underlying unstructured natural language text by detecting argumentative components (e.g., *claims* and *premises*) and the relations that connect them (Lawrence and Reed 2019; Eger, Daxenberger, and Gurevych 2017). By transforming raw text into a structured representation of argumentative content, AM makes the underlying reasoning structure explicit and amenable to systematic analysis. Political debates provide a particularly relevant and challenging setting for AM (Cabrio and Villata 2018), as they consist of dynamic, turn-based interactions in which speakers continuously introduce, support, and contest positions while referring to a variety of actors, institutions, events, and other entities. Argumentative structure alone, however, only provides a partial view of the debate. Knowing that a span expresses a claim or a premise reveals its argumentative role, but not the entities that make the argument meaningful. This distinction is particularly important in political debates, where understanding an argument often depends on identifying the people, organizations, locations, or events being discussed. Named Entity Recognition (NER) provides this complementary layer by identifying and classifying entity

mentions, yet ignores the argumentative functionalities of the statements in which they occur.

AM and NER therefore offer two complementary views of the same discourse: NER captures *what* is being discussed, while AM captures *how* statements about those entities contribute to the argumentative structure. Joint approaches can hence integrate argumentative and entity-level information within a unified representation, making explicit whether entities occur inside or outside argumentative components (La Cava, Monea, and Greco 2026).

Recent advances in Large Language Models (LLMs) have created new opportunities for such integrated analysis. Their contextual language understanding capabilities (Chang et al. 2024) have proven effective across several AM (Chen et al. 2024; Cabessa, Hernault, and Mushtaq 2025; Caputo, Greco, and La Cava 2026a) and NER (Zhou et al. 2024) sub-tasks. Yet, existing systems address only parts of the broader analysis problem. For example, ARGUS (Caputo, Greco, and La Cava 2026b) provides an LLM-based end-to-end AM pipeline with formal reasoning, but does not model the entity layer. Conversely, DISPUTool 2.0 (Goffredo et al. 2024) supports both argumentative and entity-level analysis through separate components, but does not address incremental, real-time debate analysis.

The latter is the most relevant dimension in existing analysis tool: a debate is intrinsically an evolving stream, yet argument mining systems are commonly used after a transcript or document is available. Moving from retrospective to live debate analysis changes the system requirement, rendering the lack of a temporal dimension a strong limitation. A live system must process the discussion incrementally, as new speaker turns become available, while continuously updating a coherent representation of the evolving discourse. Beyond identifying argumentative components as they appear, it should capture the entities involved in those arguments, enabling users to follow recurring actors, compare speakers’ positions, and observe how the argumentative focus evolves throughout the debate. However, existing tools do not yet provide a unified solution that combines these capabilities for real-time debate analysis.

Contribution In this paper, we fill this gap by introducing ARGUSTREAM, an interactive system that operationalizes argument mining as a *live*, *turn-synchronous*, and *joint argument-entity* analysis process. Rather than await for a

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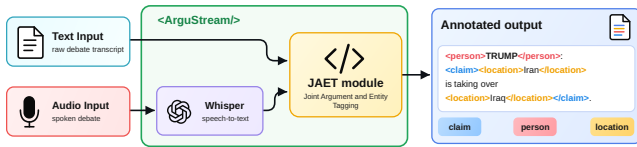


Figure 1: Overview of the ARGUStream architecture.

complete transcript, ARGUStream processes speaker turns as they unfold, and incrementally builds a unified representation of *who* is speaking, *what* is being argued, and *which entities* those arguments concern.

The ARGUStream System

Let $D = \langle u_1, u_2, \dots, u_n \rangle$ be a debate represented as an ordered stream of turns, s.t. $u_i = (s_i, x_i)$ contains the speaker metadata s_i and the corresponding textual content x_i .

Turn-level Structured Inference. For each newly available turn u_i , provided either directly as text or obtained from audio, ARGUStream produces an annotated representation $u_i^* = f(u_i)$, consisting of argumentative components and named entities identified by f . The current instantiation of f performs joint argument–entity analysis: it identifies argumentative spans and assigns them a role, e.g., *claim* or *premise*, while simultaneously detecting debate-relevant entity mentions. Both semantic layers are produced within a single inference pass and encoded as inline annotations while preserving the original turn text (La Cava, Monea, and Greco 2026). Accordingly, u_i^* exposes both the argumentative structure of the turn and the entities occurring within argumentative and non-argumentative spans.

Evolving Debate State While f operates locally on the incoming turn, ARGUStream incrementally integrates its output into a system-level representation of the discourse. Let S_i denote the debate state after processing u_i , with $S_0 = \emptyset$. Each new structured turn updates the state according to $S_i = U(S_{i-1}, u_i^*)$, where U incorporates the newly available annotation into the accumulated debate representation. This separation is crucial to ARGUStream: inference remains local to each incoming turn, while the system maintains a cumulative and explorable representation of the debate prefix observed up to time i .

Debate Exploration and Refinement. From S_i , ARGUStream derives complementary views of the evolving debate. Specifically, for each entity mention, it retains its speaker, position, type, and argumentative context, enabling the construction of a speaker-entity matrix and a temporal view of the debate. These allow users to inspect which entities are discussed by each participant, whether they occur within claims, premises, or non-argumentative text, and how the focus of the debate changes over time.

ARGUStream keeps the human in the loop by allowing users to refine automatically annotated turns through an interactive annotation environment. Users can select spans and assign argumentative or entity labels, define task-specific entity categories, and modify or remove existing annotations. Figure 1 shows overview of ARGUStream.

Live Input and Replay ARGUStream supports two complementary modes of operation that share the same inference and state-update mechanism. In a *live session*, written or spoken turns are acquired and analyzed incrementally as the debate progresses. Textual turns are directly mapped to u_i , whereas spoken ones are first associated with the corresponding speaker, and then fed into f . The system therefore does not require the complete audio or transcript before analysis can begin: annotations are appended to S_i as each processed turn becomes available, so the debate can be inspected while it is still unfolding. In *replay* mode, users can provide a complete debate, which is processed according to the same turn-by-turn workflow, thus yielding the sequence of states $S_1, S_2, \dots, S_n$, enabling the inspection of historical debates.

Implementation Details

The ARGUStream backend is exposed through a FastAPI service and supports textual, file-based, and audio input. Text is directly forwarded to the annotation module, whereas recorded speaker turns are first transcribed using Whisper (Radford et al. 2023), and the speaker is provided by the user rather than inferred through speaker-diarization. In the current implementation, the turn-level analysis function f is instantiated with JAET (La Cava, Monea, and Greco 2026), which jointly produces argument-component and debate-entity annotations in a single inference pass. The system also supports the comparison of joint and sequential inference. For the same input turn, users can inspect predictions produced by JAET alongside those obtained through AM→NER and NER→AM pipelines, making differences in span coverage, nested argument-entity annotations, and output consistency directly observable. ARGUStream further provides a manual annotation interface, allowing users to annotate textual spans and define custom entity types beyond those supported by the current implementation. Annotations can be exported in standard formats such as BIO and CoNLL for subsequent analysis and dataset construction.

Expected Impact

ARGUStream brings a concrete transition from document-level NLP analysis of argumentative debates toward systems that maintain structured interpretations of language streams as they unfold. In the challenging context of political debates, we expect ARGUStream to unlock unprecedented capabilities: by incrementally updating a shared debate state, ARGUStream supports forms of analysis that are difficult to obtain from static, post-hoc annotations alone, e.g., the evolution of argumentative focus and the comparison of entity-centered positions across participants. We expect the underlying interaction paradigm to scale across many domains, as the separation between turn-level inference and incremental state construction works with alternative annotation models and domain-specific semantic layers, making ARGUStream a useful tool for both researchers and professionals.

Release and Demonstration. ARGUStream will be made available upon acceptance and a recorded demonstration illustrating the main functionalities of the system is provided with this manuscript¹.

¹placeholder per il link

References

- Cabessa, J.; Hernault, H.; and Mushtaq, U. 2025. Argument Mining with Fine-Tuned Large Language Models. In Rambow, O.; Wanner, L.; Apidianaki, M.; Al-Khalifa, H.; Eugenio, B. D.; and Schockaert, S., eds., *Proceedings of the 31st International Conference on Computational Linguistics*, 6624–6635. Abu Dhabi, UAE: Association for Computational Linguistics.
- Cabrio, E.; and Villata, S. 2018. Five Years of Argument Mining: a Data-driven Analysis. In *Proceedings of the Twenty-Seventh International Joint Conference on Artificial Intelligence, IJCAI-18*, 5427–5433. International Joint Conferences on Artificial Intelligence Organization.
- Caputo, E.; Greco, S.; and La Cava, L. 2026a. Argument Component Segmentation with Fine-Tuned Large Language Models. In Demberg, V.; Inui, K.; and Marquez, L., eds., *Findings of the Association for Computational Linguistics: EACL 2026*, 5154–5167. Rabat, Morocco: Association for Computational Linguistics. ISBN 979-8-89176-386-9.
- Caputo, E.; Greco, S.; and La Cava, L. 2026b. ARGUS: Towards End-to-End Argument Mining with Large Language Models. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 40, 41547–41549.
- Chang, Y.; Wang, X.; Wang, J.; Wu, Y.; Yang, L.; Zhu, K.; Chen, H.; Yi, X.; Wang, C.; Wang, Y.; Ye, W.; Zhang, Y.; Chang, Y.; Yu, P. S.; Yang, Q.; and Xie, X. 2024. A Survey on Evaluation of Large Language Models. 15(3).
- Chen, G.; Cheng, L.; Luu, A. T.; and Bing, L. 2024. Exploring the Potential of Large Language Models in Computational Argumentation. In Ku, L.-W.; Martins, A.; and Srikumar, V., eds., *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 2309–2330. Bangkok, Thailand: Association for Computational Linguistics.
- Eger, S.; Daxenberger, J.; and Gurevych, I. 2017. Neural End-to-End Learning for Computational Argumentation Mining. In Barzilay, R.; and Kan, M.-Y., eds., *Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 11–22. Vancouver, Canada: Association for Computational Linguistics.
- Goffredo, P.; Cabrio, E.; Villata, S.; Haddadan, S.; and Torres Sanchez, J. 2024. DISPUTool 2.0: A Modular Architecture for Multi-Layer Argumentative Analysis of Political Debates. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 37, 16431–16433.
- La Cava, L.; Monea, S. F.; and Greco, S. 2026. Who Argues What? Joint Argument-Entity Detection and Classification in Political Debates. arXiv:2609.10192.
- Lawrence, J.; and Reed, C. 2019. Argument Mining: A Survey. *Computational Linguistics*, 45(4).
- Radford, A.; Kim, J. W.; Xu, T.; Brockman, G.; Mcleavey, C.; and Sutskever, I. 2023. Robust Speech Recognition via Large-Scale Weak Supervision. In Krause, A.; Brunskill, E.; Cho, K.; Engelhardt, B.; Sabato, S.; and Scarlett, J., eds., *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, 28492–28518. PMLR.
- Zhou, W.; Zhang, S.; Gu, Y.; Chen, M.; and Poon, H. 2024. UniversalNER: Targeted Distillation from Large Language Models for Open Named Entity Recognition. In *The Twelfth International Conference on Learning Representations, ICLR 2024*. OpenReview.net.